

MA378: Tutorial Sheet 3

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Q1. (Based on Exercise 3.1 from slides). Let $f(x) = \sin(\pi x/2)$ and let S be its PCHIP Spline interpolant at the nodes $\{x_i\}_{i=0}^3 = \{0, 1, 2, 3\}$

- (a) Evaluate S at the point $x = 2.5$.
- (b) Compute the (actual) the error at $x = 2.5$.
- (c) Compare this result with the error bound provided in Theorem 3.2

Q2. (Exer 3.2 from slides) Let $f(x) = \ln(x) - x^4$. Let l and S be the piecewise linear and Hermite cubic spline interpolants (respectively) to f on $N + 1$ equally spaced points $1 = x_0 < x_1 < \dots < x_N = 2$. What value of N would you have to take to ensure that

- (i) $\max_{1 \leq x \leq 2} |f(x) - l(x)| \leq 10^{-4}$?
- (ii) $\max_{1 \leq x \leq 2} |f(x) - S(x)| \leq 10^{-4}$?

Q3. (Based on Exer 3.3.3). There are ways of constructing the PCHIP, other than that shown in (1) of Section 3.2 above. For example, let $t = x - x_{i-1}$, then

$$S_i(x) = \frac{h^3 - 3ht^2 + 2t^3}{h^3} f_{i-1} + \frac{3ht^2 - 2t^3}{h^3} f_i + \frac{t(t-h)^2}{h^2} f'_{i-1} + \frac{t^2(t-h)}{h^2} f'_i,$$

Show that this is the same as the PCHIP.

Q4. Let $q_0, q_1, \dots, q_N$ be the quadrature weights for the Newton-Cotes rule $Q_N(f)$. For simplicity, let's assume that the quadrature rule is integrating f on the interval $[-1, 1]$. Show that $q_i = q_{N-i}$ for $i = 0, 1, \dots, N$.

Q5. Let $q_0, q_1, \dots, q_N$ be the quadrature weights for the Newton-Cotes rule $Q_N(f)$ for estimating $\int_a^b f(x) dx$.

Show that $\sum_{i=0}^N q_i = b - a$.